Keerthana K

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Education

- **Bachelor of Engineering – Biomedical Engineering:** Sri Sivasubramaniya Nadar College of Engineering
  2023–2027
  CGPA: 9.02
- **Higher Secondary (Class XII):** Velammal Vidhyashram Mambakkam (CBSE) 2023
  Percentage: 95.4%
- **Secondary School (Class X):** Velammal Vidhyashram Mambakkam (CBSE) 2021
  Percentage: 96.4%

Experience

- **Summer Intern:** May 2026 – Present
  Indian Institute of Technology (IIT) - NIID Lab
  - Working on multimodal biomedical signal and imaging analysis pipelines involving sEMG, EEG, infrared thermography, and myotonometry data for neuromuscular and physiological assessment research.
  - Building deep learning models for signal analysis and segmentation and deployable web-apps for publication.
  - Collaborating across multiple concurrent AI-for-healthcare research projects involving biomedical signal processing, medical imaging, and deep learning experimentation.
- **Data Science Intern:** May 2026 – Present
  Prediscan (Remote)
  - Working with ICD-coded healthcare datasets and cleaned patient records.
  - Contributing to disease classification, insurance cost prediction, and patient-condition analysis workflows.
  - Assisting in preprocessing, feature engineering, and machine learning pipeline development.
  - Tech: Python, SQL, Pandas, Scikit-learn, Docker

Projects

- **MediAssist AI – Clinical Report Summarizer & Medical Q&A Platform[GitHub Repository]:**
  - Developed an AI-powered web app to process medical reports, extract biomarkers using OCR and rule-based NLP pipelines, and flag abnormal lab values against reference ranges.
  - Integrated **Groq API** with **LLaMA 3.1** models to generate patient-friendly report summaries and enable conversational Q&A for uploaded reports with safety-focused responses.
  - Built an interactive dashboard with upload support, structured tables, downloadable summaries, and chatbot interface using **Streamlit**; with ongoing improvements for diverse pathology report formats.
  - **Tech: Python, Git, Streamlit, OCR (Tesseract), Regex/NLP, Groq API, LLaMA 3.1, Pandas**
- **SpaceX Launch Data Analytics and Prediction [GitHub Repository]:**
  - Collected launch datasets using **web scraping** and **SpaceX APIs** and structured them for analysis.
  - Performed exploratory data analysis using **Pandas** and **SQL** to identify launch success patterns and mission trends.
  - Developed classification models using **Decision Tree** and **Logistic Regression** to predict launch success probability.
  - Built geospatial visualizations using **Folium** and interactive dashboards to analyze launch site distribution and mission outcomes.
  - **Tech: Python, Pandas, SQL, Matplotlib, Seaborn, Plotly, Folium, Scikit-learn**
- **Hypoglycemic Tremor Detection using Wearable Sensors:**
  - Developed a wearable monitoring system using **MPU6050 IMU** and **MAX30102 PPG** sensors for hypoglycemic tremor and heart-rate detection.

- Processed sensor signals and trained a **Random Forest** model achieving **94.03% accuracy**.
- Built FastAPI-based monitoring with CSV data tracking and export.
- **Tech: Python, Statistics, Scikit-learn, FastAPI, Arduino, Embedded C, Signal Processing**

Skills

Programming: Python, SQL, Java, C (Basic)
AI / Machine Learning: Scikit-learn, TensorFlow/Keras, Supervised Learning, Deep Learning, NLP, LLM Integration, Model Evaluation
Data Science: Pandas, NumPy, Data Cleaning, Feature Engineering, EDA, Statistical analysis
Tools: GitHub, Jupyter Notebook, Tableau, PowerBI, PostgreSQL, MySQL, MATLAB
Soft skills: Leadership, Problem solving, Communication, Teamwork

Certifications

- **Data Analysis with Python (IBM):** [Certificate]
- **Databases and SQL for Data science (IBM):** [Certificate]
- **Machine Learning with Python (IBM):** [Certificate]
- **Data Visualization with Python (IBM):** [Certificate]

Extracurricular Activities

- **Design Head – IEEE EMBS SSN Chapter:** 2025–26
- **Content Editorial Member – IEEE PELS SSN Chapter:** 2025–26
- **Go For Gold Program - Accenture - Industrial exposure to Oracle Cloud, Digital marketing and Business Analytics:** 2025[Certificate]